

Maxi. Product Subarray

19.2

nums = [2, 3, -2, 4]
 0 1 2 3

Subarrays:-

{2}	{3}	{-2}	{4}
{2,3}	{3,-2}	{-2,4}	
{2,3,-2}	{3,-2,4}		
{2,3,-2,4}			

hme maxi product nikalna hai.

Teen cheeze initialize karna hoga:-

int maxprod = num[0];

int minprod = num[0];

int ans = num[0];

Phir loop :-

```
for(int i = 0; i < num.length; i++)
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```
if (num[i] < 0) {
```

```
    swap(maxprod, minprod);
```

```
}
```

→ Swap ishliye karna hoga kyunki agar min or max dono negative hoga toh may be wo ek max positive de skta hai.

ile let $-3 \times -4 = 12$ } positive
~~negative~~

dry run

nums = [2, 3, -2, 4]
0 1 2 3

i=0, num[i] = 2,

maxProd = 2

minProd = 2

ans = 2

i=1, num[i] = 3 (positive no swap)

maxProd = max(3, 2x3)
= max(3, 6)
= 6

minProd = min(3, 2x3)
= min(3, 6)
= 3

ans = max(2, 6)
= 6

i=2, num[i] = -2 (-negative swap)

Swap first! - before after
~~max~~ maxProd = 6, 3
minProd = 3, 6

maxProd = max(-2, 3x-2)
= max(-2, -6)
= -2

~~min prod~~

$$\begin{aligned}\text{min prod} &= \min(-2, 6x-2) \\ &= \min(-2, -12) \\ &= -12\end{aligned}$$

$$\begin{aligned}\text{ans} &= \max(-2, 6) \\ &= 6.\end{aligned}$$

$i=3$, $\text{num}[i] = 4$ (positive no swap).

$$\begin{aligned}\text{max prod} &= \max(4, 4x-2) \\ &= \max(4, -8) \\ &= 4.\end{aligned}$$

$$\begin{aligned}\text{min prod} &= \min(4, 4x-12) \\ &= \min(4, -48) \\ &= -48.\end{aligned}$$

$$\begin{aligned}\text{ans} &= \max(6, 4) \\ &= 6.\end{aligned}$$

Final ans.

Time Complexity $= O(N)$.

Space Complexity $= O(1)$